Is That for Me? Addressee-Aware Turn-Keeping for Open Full-Duplex Speech Models under Same-Voice Side-Talk

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Abstract

Open full-duplex speech models treat every voice in the user channel as meant for them, so on three recent benchmarks they answer when the user turns to a third person. The shipped fix in Gemini 3.8 Live is unpublished, and open remedies gate on speaker identity, which fails when the side-talk is in the user's own voice. We measure how much identity gating solves, build a de-shortcut same-voice test set with voice cloning, LoRA-fine-tune Moshi on cue-controlled synthetic side-talk, and ablate lexical against acoustic cues.

1 Problem and gap

A full-duplex model must decide, frame by frame, whether incoming speech is for it. A 2026 survey (Lu et al., 2026) lists "third-party / not-directed speech" as an ontology cell that speaker-embedding and relevance gates target but have "not shown solved". On Full-Duplex-Bench (FDB) v1.5 (Lin et al., 2025), when the user talks to someone else, GPT-4o Realtime (OpenAI, 2024) responds on 91% of clips and Freeze-Omni (Wang et al., 2024) on 58%; Moshi responds on 20% and is unclassifiable on another 59%, and the authors conclude that systems "must strengthen addressee detection". The HumDial challenge (Wang et al., 2026) and MSI-Bench (Xiong et al., 2026) report the same failure: models frequently respond to speech that was not addressed to them. Gemini 3.8 Live ships always-on "proactive audio" that "can proactively decide not to respond" (Google, 2026), and Realtime-Venus (Ant Group, 2026) reports 88% correct ignoring on this cell; neither releases weights or a method.

Every open remedy gates on *who* is speaking: FireRedChat's personalised VAD (Chen et al., 2025; Ding et al., 2019) and IRAF's end-to-end ECAPA gate (Zhong et al., 2026). Yet FDB v1.5 deliberately separates two cells: *bkg_speech* is a different voice at -15 dB, whereas *talking_other* is the *user's own voice* at -8 dB with far-field reflections, saying something meant for a third person. Identity gating can only address the first. To our knowledge, no public model has been trained on same-voice side-talk and no corpus labels it at scale (HumDial's training set has zero such items).

2 Research questions and hypotheses

RQ1 Which part of the not-addressed cell does identity gating solve? *H1:* a verification gate mutes most different-voice speech but passes most same-voice side-talk. **RQ2** Can a Moshi-class model learn to keep its turn under same-voice side-talk while still yielding to genuine interruptions? *H2:* LoRA on cue-controlled synthetic side-talk at least halves base Moshi's *side-talk onset rate* (share of side-talk events after which the model starts speaking, by VAD) at ≤ 5 points more missed interruptions. **RQ3** Which cue does it learn? *H3:* with acoustic cues (level drop, far-field colouring) present, the gain survives removing the lexical cue but not the acoustic one.

3 Proposed system

Base and baselines. Moshiko-7B (Défossez et al., 2024) (CC-BY 4.0, official `moshi-finetune` LoRA) is the fine-tuning target; PersonaPlex-7B (Roy et al., 2026) and Freeze-Omni are inference baselines. Two gates over the same Moshi aim at side-talk directly: a vocative rule (the pilot's comma-within-three-words rule) and a cascade addressee classifier (streaming ASR plus a local 8B LLM judging whether each utterance addresses the system). **G0 identity gate (no training).** ECAPA (Desplanques et al., 2020) cosine between each 0.5–1 s user-channel window and an enrollment from the user's earlier speech; windows below a held-out-tuned threshold are zeroed before the model hears them. **Data pipeline.** Behavior-SD (Lee

et al., 2025) (108k synthetic dialogues, CC-BY 4.0) supplies base dialogues. An LLM writes side-talk addressed to a named third party (with and without a vocative) and genuine interruptions; Chatterbox (Resemble AI, 2025) voice cloning renders them in the user's voice; they enter the user channel under three acoustic conditions: FDB-matched far-field (-8 dB, high-shelf cut, reflections), near-field (lexical cue only) and content-neutral far-field (acoustic cue only). Different-voice background speech (LibriSpeech (Panayotov et al., 2015), 0–15 dB SNR) is the control cell. **Training target.** Moshi keeps producing its utterance through non-addressed speech and yields on addressed interruptions; LoRA rank 128, about 2,000 steps, one H100 ($\approx$40 GB).

4 Experimental design

Test sets. (i) FDB v1.5: talking_other, bkg_speech, backchannel, interruption (498 clips). (ii) HumDial-FDBench test set (600 third-party and 200 talking-to-others items, human actors). (iii) Our de-shortcut set: same vs. different voice $\times$ acoustic cue $\times$ vocative, on unseen voices, with far-field genuine interruptions and "Hey Moshi, ..." vocatives aimed at the system. (iv) 50–100 clips recorded by the four authors (no outside participants). (v) Multi-turn sessions: 200 held-out Behavior-SD dialogues of 1–2 min, each with 2–4 same-voice side-talk events and 1–2 genuine interruptions. **Metrics** (conditions: Moshi, +vocative rule, +ASR-LLM gate, +G0, +LoRA, +G0+LoRA, PersonaPlex, Freeze-Omni)**.** Side-talk onset rate (primary), with FDB respond labels alongside; response rate and stop latency on genuine interruptions; content quality (FDB judge); gate latency within the 80 ms frame. On set (v), per-session scores: share of sessions with every event handled correctly, cumulative spurious responses, and resumption (LLM-judged: does the model return to its prior topic after side-talk). **Estimands and analysis.** Primary: relative change in side-talk onset rate (LoRA vs. base) with the interruption miss rate held within 5 points. Secondary: G0's split between different-voice and same-voice speech. Paired bootstrap CIs by clip (by session on set v); three LoRA seeds; the 2×2 cue ablation on set (iii) answers RQ3.

5 Pilot evidence

On FDB v1.5 (498 clips, ECAPA) the identity cue behaves as H1 predicts (Table 1): the gate separates background speech from real interruptions almost perfectly (AUC 0.9998) yet accepts 88–92% of same-voice side-talk at thresholds that keep 95% of the user's own speech. The pilot also exposed two shortcuts: 99 of 100 side-talk clips open with a vocative, so a comma-within-three-words rule flags 99% at 3% false positives, and all side-talk is far-field while all interruptions are near-field. Caveat: interruptions use many voices but the other cells share one TTS voice; matched on voice, ECAPA separates clean user speech from side-talk at AUC only 0.61, another reason for set (iii).

FDB v1.5 scenario	cos.	pass	muted
user_interruption (keep)	.49	.995	.13
talking_other (want mute)	.39	**.89**	**.27**
user_backchannel (keep)	.19	.52	.47
bkg_speech (mute)	.06	.01	.83

Table 1: Identity cue on FDB v1.5: median ECAPA cosine to the user; clips accepted as the user at the EER threshold; audio muted by the 0.5 s G0 gate.

6 Plan, roles and feasibility

Week 1 (to 11/12): baselines and side-talk gates on FDB v1.5 and HumDial, G0, LoRA smoke test, synthesis v0. Milestone 11/19: baseline table with the G0 split, first LoRA. Weeks 3–4: LoRA sweep over cue conditions and data size, de-shortcut, human and multi-turn sets, latency. Week 5: poster (12/3), paper (12/8). **Roles.** Zimo Fan: evaluation and baselines; Ruihuan Gao: synthesis and test sets; Ruize Sun: G0 and analysis; Weiyi Sun: LoRA. **Compute (PACE-ICE, estimated).** Real-time inference on an L40S ($\approx$24 GB): $\sim$15 h of audio per condition $\times$ $\sim$12 runs $\approx$ 180 GPU-h. LoRA on one H100: $\approx$3 h $\times$ $\sim$20 runs $\approx$ 60 GPU-h. API (GPT-4o FDB judge, resumption judge, side-talk generation): about \$150.

7 Risks and fallback

(1) The model may learn "quieter means ignore" rather than addressee semantics; the cue ablation turns this into a finding. (2) `moshi-finetune` is validated only on Moshiko; a week-1 smoke test gates the rest. (3) Synthetic side-talk may not transfer to real rooms; HumDial's human recordings and our recorded set bound the gap. If LoRA fails, we deliver a measurement paper: all non-LoRA conditions on the five test sets, the identity-gate split and the de-shortcut analysis.

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